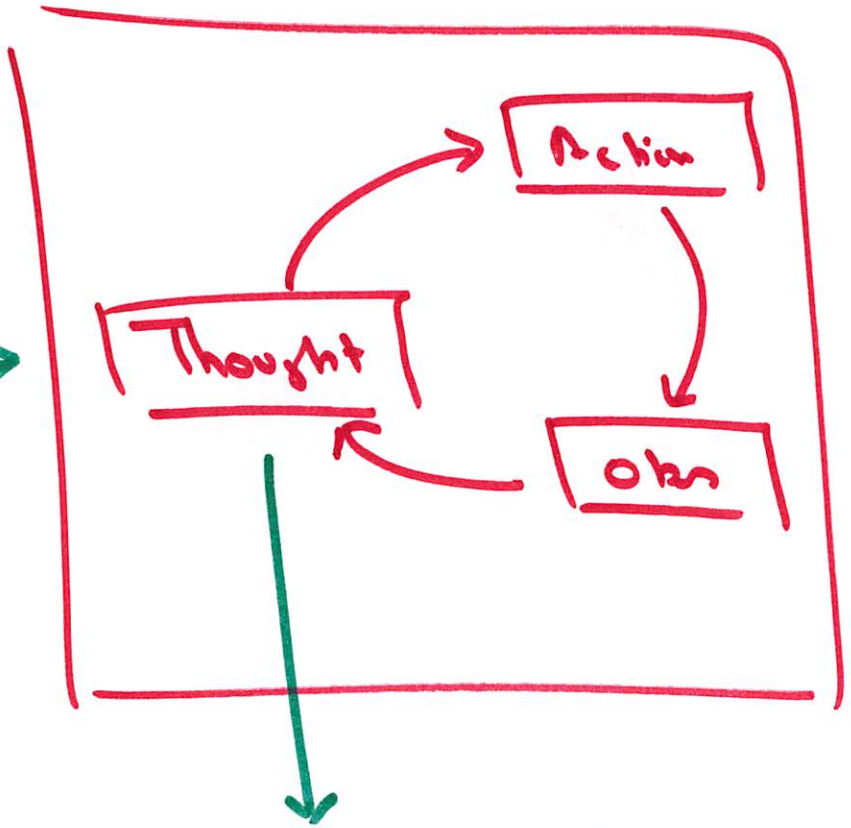
What is the birth city of the discoverer of Radium



Radium was discovered by Marie Curie and she was born in Warsaw

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Reasoning = Planning + Acting

- Reasoning is not just thinking!
  - Goal-directed planning
  - Task decomposition (eg. Plan and book a 3-day trip to Tokyo)
  - State tracking (what have I done? what is left to do?)
  - Adaptive Execution – The capacity to pause, observe real-world feedback, and adjust the plan (The Feedback Loop)
- Need carefully devised prompts that,
  1. Define the Persona & Objective (The Goal)
  2. Expose Available Tools/APIs (The Action Space)
  3. Inject the "Scratchpad" of history (The State/Memory)
  4. Enforce the Thought-Action-Observation format (The Loop)

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The ReAct Framework

```
Answer the following questions as best you can. You are a helpful assistant focused on accuracy.

You have access to the following tools:
{tools}

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of {tools}
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can repeat N times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Begin!

Question: {input}
Thought: {agent_scratchpad}
```

The ReAct Prompt (Yao et al., 2022, LangChain)

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Plan-and-Execute vs ReAct

- ReAct: You figure it out step-by-step. Good for high uncertainty. (Slow, Expensive, Flexible)
- Plan-and-Execute: Make the full list of steps before you start. Good for low uncertainty. (Rigid, Efficient, Fragile)

```
You are an expert Planner.
Your goal is to understand the user's request and create a step-by-step
plan to achieve it.
The plan should be a list of actionable steps.
Do not execute the steps yourself. Just list them.

User Request: {input}

Output Format:
1. Step 1
2. Step 2
3. ...
```

```
You are an expert Executor.

Your goal is to execute the current step of the plan.

# THE PLAN
{plan}

# RELEVANT CONTEXT
The following steps have already been completed:
{previous_steps}

# YOUR CURRENT TASK
Step: {current_step}

Execute this step and return the result.
```

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Wang et al. 2023. LangChain

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"Pick a lunch spot"

- ReAct: Search → see options → measure walking time → fetch menu → choose → email.
  Iterates with evidence each step; needs a recursion_limit.
- Plan-and-Execute: Plan: (1) find candidates, (2) filter by distance, (3) check price, (4) email. Then execute steps; re-plan if a step fails.
- Tool-calling agent: "Find a cheap lunch near me." → calls Search API once → returns a guess.
  If wrong, you must ask again; no self-correction.
- AutoGPT-style: Sets sub-goals, saves notes to memory, tries different searches, maybe overthinks; needs strict budget/step caps.
- LangGraph DAGs: Nodes: search → filter_distance → check_price → decision → send_email; add edges for errors/timeouts; swap a node without changing the rest.

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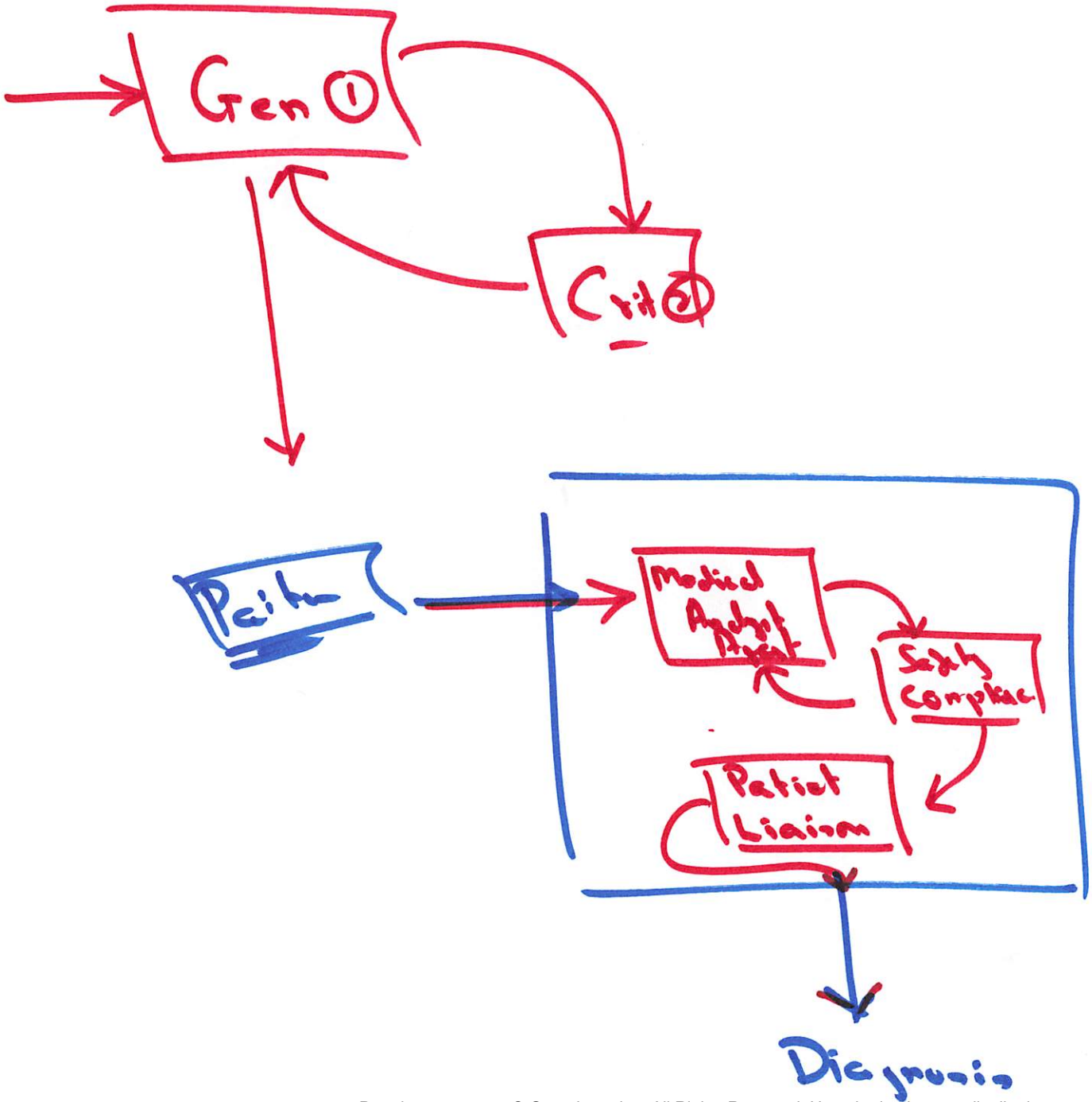
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System 1 vs. System 2

- Daniel Kahneman – Thinking, Fast and Slow
  - System 1: Fast / Intuitive, Effortless, always active
  - System 2: Slow / Deliberate, Effortful, requires attention and focus
- Self-Reflection & Critique
- Agents can hallucinate
  - 1 Generate–Evaluate–Refine loop
  - 2 Critic agents improve quality

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